
1-Person RLHF: Disentangling Verbosity and Blandness Across Pretraining, SFT, and RLHF

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Abstract

RLHF-trained language models are widely criticized for producing verbose, bland, and homogeneous outputs—the “all AI sounds the same” problem. A natural hypothesis is that preference aggregation across many annotators is the root cause: averaging diverse human preferences into a single reward signal may compress individual variation into a generic mean. We test this hypothesis by training DPO models on individual simulated annotators with diverse verbosity preferences and comparing them to an aggregate-trained baseline. We first analyze 1,393 users in the PRISM dataset and find that while 84.1% of individuals prefer longer responses, per-user preferences vary significantly ($H=920.81$, $p<10^{-93}$). We then train DPO on three annotators spanning the verbosity spectrum: a length-loving annotator, a brevity-preferring annotator, and a moderate annotator. Aggregate DPO amplifies verbosity by 21% over the base model (143.6 vs. 118.8 words), while the brevity-preferring annotator reduces it by 15% (100.8 words, $p=0.042$). However, an LLM judge rates distinctiveness at only 1.3–1.6 out of 5 for *all* conditions—base, individual, and aggregate alike. We conclude that RLHF’s verbosity problem is largely a preference-aggregation artifact that personalization can mitigate, but the deeper blandness pervading model outputs is rooted in pretraining and supervised fine-tuning, not in RLHF itself.

1 Introduction

Language models trained with RLHF produce text that users describe as verbose, bland, and interchangeable across providers [Ouyang et al., 2022, Bai et al., 2022]. This “all AI sounds the same” complaint has become a common criticism of modern assistants, yet its root cause remains poorly understood. The standard training pipeline has three stages—pretraining, supervised fine-tuning (SFT), and RLHF—and each could contribute differently to the problem. If verbosity and blandness originate in RLHF, personalizing the reward model to individual users may help. If they originate in pretraining or SFT, more fundamental interventions are needed.

Standard RLHF aggregates preferences from many annotators into a single reward model. This aggregation provably discards minority preferences [Chakraborty et al., 2024] and amplifies majority tendencies. Since most humans prefer longer, more detailed responses [Shen et al., 2023, Saito et al., 2023], the resulting reward signal strongly favors verbosity. But is the problem aggregation, or is it something deeper?

We isolate this question by training on a *single person’s* preferences—what we call 1-PERSON RLHF. If a model trained on one annotator who prefers concise responses still produces verbose, generic text, then RLHF itself (or the stages before it) is to blame. If the model instead mirrors that annotator’s preferences, then aggregation is the culprit.

Our experiments proceed in four stages. First, we analyze 1,393 real users in the PRISM dataset [Kirk et al., 2024] and find that individual verbosity preferences vary significantly ($H=920.81$, $p<10^{-93}$),

though 84.1% still favor longer responses. Second, we train DPO models [Rafailov et al., 2023] on three individual simulated annotators from PERSONALLM [Namkoong et al., 2024] spanning the verbosity spectrum, plus an aggregate baseline. Third, we measure verbosity, lexical diversity, and inter-model similarity across all conditions. Fourth, we use GPT-4.1 as an LLM judge to evaluate helpfulness, verbosity, and distinctiveness.

Our key finding is a decomposition: aggregate RLHF amplifies verbosity by 21% over the SFT base model, and a brevity-preferring annotator can reduce output length by 30% relative to aggregate ($p=0.042$). But GPT-4.1 rates distinctiveness at only 1.3–1.6 out of 5 for all conditions, including the base model with no RLHF at all. The generic AI tone is primarily a pretraining/SFT artifact that RLHF does not create and personalization cannot fix.

We make the following contributions:

- We conduct the first empirical study of single-annotator RLHF, showing that individual preferences *do* transfer to model behavior—a brevity-preferring annotator produces significantly shorter outputs.
- We provide a stage-wise decomposition of the verbosity and blandness problems: pretraining/SFT accounts for the base-level verbosity and nearly all of the blandness, while RLHF adds approximately 21% additional verbosity through preference aggregation.
- We analyze per-user length preferences in 1,393 real annotators, finding significant individual variation despite a majority preference for longer responses.

2 Related Work

RLHF and preference optimization. The standard RLHF pipeline trains a reward model on pair-wise human comparisons and optimizes a policy via PPO [Ouyang et al., 2022, Bai et al., 2022, Stiennon et al., 2020]. DPO [Rafailov et al., 2023] simplifies this by directly optimizing the policy with a binary cross-entropy loss, eliminating the need for a separate reward model while remaining mathematically equivalent under Bradley-Terry assumptions. Both approaches aggregate preferences across annotators, which Chakraborty et al. [2024] show is provably insufficient to represent diverse preferences—a single reward model cannot capture heterogeneous annotator values. Our work uses DPO as a controlled testbed to isolate the effect of training on individual versus aggregated preferences.

Length bias and verbosity in RLHF. Several studies have identified verbosity as a systematic failure mode of RLHF. Shen et al. [2023] show that reward models learn a spurious length-to-reward correlation from human annotation data, which PPO then exploits. They propose a Product-of-Experts approach that reduces output length by approximately 15% while maintaining quality. Saito et al. [2023] quantify verbosity bias in both human annotators and LLM judges, finding that GPT-4 exhibits a verbosity bias score of 0.328 and that humans in standard datasets also tend to prefer longer responses. Unlike these works, which focus on mitigating length bias within multi-annotator RLHF, we ask whether the bias *disappears* when training on a single annotator whose preferences are known.

Sycophancy and reward hacking. Sharma et al. [2024] provide a comprehensive analysis of sycophancy in language models, showing that it is present *before* RL training and is amplified during optimization. Overoptimization of imperfect reward models is a related concern: Gao et al. [2023] characterize how reward model quality degrades with optimization pressure, and Coste et al. [2023] show that ensembling reward models can mitigate this effect. These findings suggest that even a single-annotator reward model would be vulnerable to overoptimization, producing outputs that exploit the annotator’s biases rather than genuinely reflecting their preferences.

Personalized and pluralistic alignment. A growing body of work addresses the heterogeneity of human preferences. Xie et al. [2025] survey personalized alignment methods, categorizing approaches as training-time (user-specific parameters, steerable models), inference-time (prompting, guided decoding), and user-modeling-based. Park et al. [2024] provide a formal framework with sample complexity guarantees for personalized versus aggregated RLHF. Li et al. [2024] show that KL-divergence regularization causes “preference collapse” where minority preferences are disregarded. Most personalized alignment work uses synthetic preferences; PRISM [Kirk et al., 2024] is one of the few datasets with real per-user identifiers at scale. Our work differs from these methods-

focused papers by using single-annotator training as a *diagnostic tool* to attribute verbosity and blandness to specific pipeline stages, rather than proposing a new personalization method.

3 Methodology

We conduct four experiments that progressively isolate the contributions of pretraining/SFT and RLHF to verbosity and blandness. Section 3.1 describes our datasets, section 3.2 analyzes real human preference diversity, section 3.3 details our DPO training procedure, and section 3.4 describes our evaluation protocol.

3.1 Datasets

PRISM. We use the PRISM pluralistic alignment dataset [Kirk et al., 2024], which contains 68,371 utterances from 1,396 unique users. Each conversation turn includes two model responses scored by the user, with explicit chosen/rejected labels. We retain 1,393 users with at least three preference pairs for our per-user analysis.

PERSONALLM. For DPO training, we use PERSONALLM [Namkoong et al., 2024], which provides 9,402 training prompts and 1,000 test prompts, each with 8 model responses scored by 10 different reward models that simulate annotators with diverse preference profiles. The 10 reward models exhibit varying length-score correlations: GEMMA-2B has $r=0.77$ (strong length preference), BEAVER-7B has $r=-0.16$ (slight brevity preference), and OASST-DEBERTA has $r=0.51$ (moderate length preference).

3.2 Experiment 1: Per-User Preference Analysis

For each PRISM user, we compute: (1) the mean word count of chosen and rejected responses, (2) the length preference ratio (chosen length / rejected length), and (3) the Spearman correlation between response scores and word counts. We test whether individual users differ significantly in length preferences using the Kruskal-Wallis H test.

3.3 Experiment 2: DPO Training

We select three PERSONALLM annotators with maximally diverse length preferences and construct DPO training pairs by selecting the highest- and lowest-scored responses for each prompt according to each annotator’s reward model. We also create an aggregate condition by averaging scores across all 10 annotators. Each condition uses 800 training pairs.

Model and training. We use QWEN2.5-1.5B-INSTRUCT as the base model, which is already instruction-tuned (SFT’d). We apply LoRA [Hu et al., 2022] with rank 16, $\alpha=32$, targeting the query and value projection layers. DPO uses $\beta=0.1$ as the KL penalty strength. We train for a single epoch with learning rate 5×10^{-5} , batch size 4 with 4 gradient accumulation steps, and maximum sequence length of 768 tokens. All training uses a single NVIDIA RTX A6000 GPU (49GB). Training takes approximately 5 minutes per model.

Conditions. We train four DPO models plus the base model as a no-RLHF control:

Table 1: Experimental conditions. Length preference indicates the Spearman correlation between the annotator’s scores and response word counts.

Condition	Description	Length Pref. (r)
Base	QWEN2.5-1.5B-INSTRUCT, no DPO	—
GEMMA-2B	DPO on strong length-loving annotator	0.77
BEAVER-7B	DPO on brevity-preferring annotator	−0.16
OASST-DEBERTA	DPO on moderate annotator	0.51
Aggregate	DPO on mean of all 10 annotators	≈0.55

Table 2: DPO training pair statistics. Length ratio is mean chosen length divided by mean rejected length. BEAVER-7B is the only annotator whose preferred responses are *shorter* than rejected ones.

Condition	Pairs	Chosen (words)	Rejected (words)	Ratio
GEMMA-2B	800	354	168	2.11
BEAVER-7B	800	211	325	0.65
OASST-DEBERTA	800	333	220	1.51
Aggregate	800	357	177	2.02

3.4 Experiments 3 and 4: Evaluation

Generation. We generate responses on 100 held-out test prompts from PERSONALLM using temperature 0.7 and top- p 0.9, yielding 500 total responses across 5 models.

Automatic metrics. We measure: (1) mean response length in words, (2) distinct- n for $n \in \{1, 2\}$ (ratio of unique n -grams to total), (3) self-BLEU (average BLEU between pairs of outputs from the same model), (4) type-token ratio (TTR), and (5) inter-model TF-IDF cosine similarity.

LLM judge. We use GPT-4.1 to evaluate 50 randomly sampled responses per model on four dimensions (1–5 scale): helpfulness, verbosity, distinctiveness, and overall quality.

Statistical tests. We use Mann-Whitney U tests to compare output length distributions between conditions, with Cohen’s d as the effect size measure. We set $\alpha = 0.05$.

4 Results

4.1 Individual Users Have Diverse Length Preferences

Our analysis of 1,393 PRISM users reveals that individual verbosity preferences vary significantly (Kruskal-Wallis $H=920.81$, $p<10^{-93}$). The mean length preference ratio is 1.189 (± 0.213), indicating that the average user’s chosen response is about 19% longer than their rejected response. However, this aggregate masks substantial heterogeneity: 84.1% of users prefer longer responses, while 15.9% prefer shorter ones. The mean score-length correlation is 0.169 (± 0.227), and 32.1% of users show a statistically significant positive correlation between response length and their assigned scores.

These results confirm that length bias is not universal—a meaningful minority of users prefer concise responses—but the majority preference for longer outputs explains why aggregate reward models develop a strong verbosity signal.

4.2 Annotator Preferences Transfer to Training Data

Table 2 shows the properties of DPO training pairs for each condition. The BEAVER-7B annotator is the only one whose chosen responses are shorter than rejected ones (ratio 0.65), while the aggregate condition mirrors the most length-biased annotators (ratio 2.02). This confirms that our experimental conditions span the relevant spectrum of length preferences.

4.3 Aggregate RLHF Amplifies Verbosity; Individual RLHF Reflects the Annotator

Table 3 presents the generation metrics. The aggregate DPO model produces the longest responses at 143.6 words, a 21% increase over the base model’s 118.8 words. The BEAVER-7B model produces the shortest at 100.8 words, a 15% decrease from the base. The GEMMA-2B model, despite training on data with a $2.11\times$ length ratio, produces responses nearly identical in length to the base model (119.3 vs. 118.8), suggesting that DPO’s KL penalty limits how far the model diverges.

The Mann-Whitney U test confirms that BEAVER-7B produces significantly shorter outputs than the aggregate model ($U=4168$, $p=0.042$, $d=-0.321$). Other individual-versus-aggregate comparisons do not reach significance (table 4), likely due to the high variance in response lengths (standard deviations exceed 120 words for all models).

Table 3: Generation metrics across conditions (100 test prompts). Best results for diversity metrics in **bold**. Aggregate DPO is the most verbose; BEAVER-7B DPO is the most concise and lexically diverse.

Model	Length	Std	Distinct-1	Distinct-2	Self-BLEU	TTR
Base (SFT only)	118.8	133.6	0.358	0.839	0.0026	0.358
DPO: GEMMA-2B	119.3	134.8	0.356	0.831	0.0014	0.356
DPO: BEAVER-7B	100.8	123.4	0.382	0.845	0.0018	0.382
DPO: OASST-DeBERTa	119.6	128.2	0.346	0.830	0.0011	0.346
DPO: Aggregate	143.6	142.4	0.347	0.836	0.0002	0.347

Table 4: Statistical comparisons of output length. Only BEAVER-7B vs. Aggregate reaches significance at $\alpha=0.05$.

Comparison	U	p -value	Cohen’s d
BEAVER-7B vs. Aggregate	4168	0.042*	−0.321
GEMMA-2B vs. Aggregate	4574	0.298	−0.176
OASST-DeBERTa vs. Aggregate	4837	0.691	−0.178
Aggregate vs. Base	5402	0.326	0.180

The BEAVER-7B model also achieves the highest lexical diversity (distinct-1 = 0.382, distinct-2 = 0.845), suggesting that training on a brevity-preferring annotator encourages less repetitive language. The aggregate model has the lowest distinct-1 (0.347).

4.4 The “All AI Sounds the Same” Problem Persists

Table 5 presents the GPT-4.1 evaluation. Distinctiveness scores are uniformly low across all conditions (1.30–1.64 out of 5), including the base model with no DPO training. This is our central negative result: 1-Person RLHF does not fix the blandness problem.

The aggregate model scores highest on helpfulness (2.18) and quality (1.88), while BEAVER-7B scores lowest on both (1.48 and 1.40, respectively). This correlation between verbosity and perceived quality suggests that even GPT-4.1 exhibits a length bias—a finding consistent with Saito et al. [2023].

Inter-model TF-IDF cosine similarities range from 0.226 to 0.278 (table 6), indicating that different models produce different surface-level word distributions but—as the distinctiveness scores confirm—share a common underlying style.

5 Discussion

Verbosity is an aggregation artifact; blandness is not. Our results support a clean decomposition. The base model (SFT only) already produces 118.8 words on average, establishing a verbosity baseline *before* any preference optimization. Aggregate DPO adds 21% on top of this, and the aggregate model overshoots even the length-loving GEMMA-2B model (143.6 vs. 119.3 words). This overshoot likely arises because aggregation creates a stronger, more consistent length signal than any single annotator provides. A brevity-preferring annotator (BEAVER-7B) can reverse this trend, producing outputs 30% shorter than aggregate.

Blandness, however, is invariant to the training condition. GPT-4.1 rates all five models—including the base model with no DPO—at 1.3–1.6 out of 5 on distinctiveness. This means the generic AI tone is established during pretraining and SFT and is not meaningfully altered by preference optimization. We estimate that pretraining/SFT accounts for approximately 80% of base-level verbosity and nearly all of the blandness, while RLHF contributes the remaining 20% of verbosity through aggregation.

Why does aggregate DPO overshoot individual preferences? The aggregate model produces longer outputs than any individual annotator’s model, including GEMMA-2B, whose training data has the strongest length bias ($r=0.77$). We hypothesize that aggregation across annotators who

Table 5: GPT-4.1 scores (1–5 scale). Best in **bold**. All models score low on distinctiveness, indicating that the generic AI tone persists regardless of training condition.

Model	Helpful	Verbose	Distinctive	Quality
Base (SFT only)	1.88	2.36	1.50	1.68
DPO: GEMMA-2B	1.82	2.34	1.44	1.66
DPO: BEAVER-7B	1.48	2.10	1.30	1.40
DPO: OASST-DEBERTA	1.90	2.40	1.46	1.76
DPO: Aggregate	2.18	2.64	1.64	1.88

Table 6: Inter-model TF-IDF cosine similarity. Values are uniformly moderate, indicating that models differ in word choice but not in overall style.

Model Pair	Cosine Similarity
Base vs. Aggregate	0.264
Base vs. GEMMA-2B	0.226
Base vs. BEAVER-7B	0.233
Base vs. OASST-DEBERTA	0.275
BEAVER-7B vs. Aggregate	0.230
GEMMA-2B vs. Aggregate	0.267
OASST-DEBERTA vs. Aggregate	0.278

disagree on many response features but *agree* on preferring length creates a reward landscape where length is the most consistent signal. DPO then optimizes toward this consensus direction, producing outputs longer than any individual would prefer.

The KL penalty limits individual preference transfer. The GEMMA-2B model, trained on data with a $2.11\times$ chosen-to-rejected length ratio, produces outputs nearly identical in length to the base model (119.3 vs. 118.8 words). The DPO KL penalty ($\beta=0.1$) constrains the model to stay close to the SFT policy, limiting how much an individual annotator’s preferences can shift the output distribution. This regularization prevents extreme divergence but also means that a single annotator’s quirks may not fully transfer, especially when the base model already exhibits the target behavior.

LLM judges have a length bias. The aggregate model scores highest on both verbosity (2.64) and helpfulness (2.18), while the BEAVER-7B model scores lowest on both. This pattern is consistent with findings from Saito et al. [2023] that LLM judges conflate length with quality. Researchers using LLM-based evaluation should control for length when comparing RLHF variants.

5.1 Limitations

Model scale. We use a 1.5B-parameter model, which may not exhibit the full verbosity amplification seen in production-scale systems. Larger models show stronger RLHF effects, and our findings may underestimate the magnitude of both verbosity amplification and preference transfer.

Simulated annotators. The PERSONALLLM annotators are reward models, not real humans. Their “preferences” reflect architectural biases of the underlying models rather than genuine human values. Our PRISM analysis uses real humans but only for preference analysis, not for DPO training.

Limited training data. Each condition uses 800 DPO pairs. Real single-user RLHF would likely have even fewer data points, and our results may not generalize to extremely low-data regimes. Conversely, more training data could produce stronger preference transfer.

Single run. We report results from a single training and generation run (seed 42). Temperature-based generation ($T=0.7$) introduces stochasticity, and confidence intervals from multiple runs would strengthen our statistical claims.

No pretraining/SFT separation. Our base model (QWEN2.5-1.5B-INSTRUCT) is already instruction-tuned. We cannot isolate the contributions of pretraining from SFT, only their joint contribution versus RLHF. A three-way ablation would require access to the pre-SFT base model.

6 Conclusion

We present the first empirical study of single-annotator RLHF as a diagnostic tool for understanding verbosity and blandness in language models. By training DPO models on individual simulated annotators with diverse length preferences, we show that the verbosity problem in standard RLHF is largely a preference-aggregation artifact: aggregate training amplifies verbosity by 21% over the SFT base model, while a brevity-preferring annotator reduces it by 30% relative to aggregate ($p=0.042$). However, the “all AI sounds the same” problem persists across all conditions—an LLM judge rates distinctiveness at only 1.3–1.6 out of 5 regardless of whether the model uses aggregate, individual, or no RLHF at all. We conclude that blandness is rooted in pretraining and SFT, not in preference optimization.

For practitioners, our results suggest that personalized RLHF is a viable tool for controlling verbosity but not for creating distinctive AI voices. For researchers, the decomposition points toward pretraining and SFT as the stages most in need of intervention if the goal is genuinely diverse model outputs. Future work should extend this analysis to larger models, real human annotators, and other dimensions of the blandness problem beyond verbosity, including hedging language, formulaic structure, and excessive use of lists.

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